CS341 · COMPUTER GRAPHICS

Question Bank Solutions

Practice on every midterm chapter

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01 Solution A: true / false

#	Answer	Reason
1	**True**	Display list = refresh display file.
2	**False**	That is one of its disadvantages.
3	**True**	Along with the laser light show.
4	**True**	TV technology.
5	**True**	Dot matrix and laser.
6	**True**	0, 2, 4, then 1, 3, 5.
7	**False**	307,200 **bits** = 38,400 bytes.
8	**True**	
9	**False**	Low persistence suits animation; high persistence suits complex static pictures.
10	**False**	LCD is nonemissive.
11	**False**	That is active matrix. Passive: one transistor per row or column.
12	**True**	
13	**True**	
14	**True**	`0.3 + 0.7 = 1`
15	**False**	3D only.
16	**False**	The reverse: every convex combination is affine.

17	**True**	
18	**True**	
19	**True**	
20	**False**	DDA uses floating point, then rounds.

02 Solution B: multiple choice

#	Answer	Reason
1	**b**	vector, stroke-writing, calligraphic
2	**b**	The frame buffer belongs to raster.
3	**c**	
4	**c**	A joystick is an input device.
5	**a**	
6	**c**	
7	**a**	
8	**b**	$\sqrt{(9 + 16)} = 5$
9	**b**	$a\perp = (-ay, ax) = (-5, 2)$
10	**a**	A point has 1.
11	**c**	$1 \cdot 2 + 2 \cdot (-1) = 0$
12	**c**	A hard-copy device.
13	**b**	$\frac{\frac{1080}{5}}{\frac{1920}{10}} = \frac{216}{192} = 1.125$
14	**b**	$\text{steps} = \max(\lvert 3 \rvert, \lvert 6 \rvert) = 6$
15	**c**	

03 Solution C: fill in the blanks

1. scan line 2. pixel 3. refresh rate 4. dot pitch 5. frame buffer 6. display list
7. Q – P 8. one-tenth 9. emissive · nonemissive 10. p5
11. glu · glut 12. the larger (max)

04 Solution D: short answers

1. Random vs raster

	Random	Raster
Storage	Display list (line commands)	Frame buffer (pixel values)
Resolution	Very high, smooth lines	Stair-step lines
Color / shading	Limited, no realistic shading	Easy

2. Random scan

Advantages: very high resolution; easy animation; little memory.
Disadvantages: needs an intelligent, processor-controlled beam; flicker limits screen density; can't draw complex images; limited and expensive color.

3. CRT parts

Connector pins (base), electron gun, focusing system, vertical and horizontal deflection plates, phosphor-coated screen, electron beam.
The deflection plates steer the beam so it can hit any spot on the screen.

4. Beam-penetration vs shadow-mask

Beam-penetration: layers of colored phosphor; the color depends on how far the beam penetrates.
Shadow-mask: three electron guns (R, G, B) and a mask whose holes line up with triangles of red, green and blue phosphor dots.

5. Active vs passive matrix

Active: one transistor per pixel; brighter and sharper; more complicated and expensive.
Passive: one transistor per row or column; cheaper; needs less power.

6. Aspect ratio

To maintain geometric features of figures: a square stays a square and a circle does not become an ellipse.

7. DDA problems

1. Rounding of the floating-point values makes pixels drift away from the true line path for long line segments.
2. Rounding operations are time consuming.

8. Absolute vs relative

Absolute: the actual position in the coordinate system.
Relative: an offset from the current (last) position, e.g. $(3, 8) + (2, -1) \rightarrow (5, 7)$.